

Monetary Transmission to Prices: Disaggregated Evidence from Norway*

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Abstract

We investigate how monetary policy affects disaggregated consumer prices in Norway. There are three main findings. First, prices tend to increase less in response to a higher interest rate, but the response is delayed and uncertain. Second, the average response hides dispersion. Following a higher interest rate, some prices grow less, most prominently *accommodation services* and *food and non-alcoholic beverages*, while other prices grow faster, such as *rentals for housing*. Third, we explore the cost, aggregate demand, and exchange rate channels of monetary policy. We find the clearest evidence for the cost channel: sectors with high debt as a share of total costs raise prices in response to higher interest rates. Our results are also consistent with the aggregate demand and exchange rate channels, but the estimates for these two channels are imprecise and remain inconclusive.

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1 Introduction

What is the effect of higher interest rates on consumer prices? And through which mechanisms does this effect operate? For a central bank charged with maintaining price stability, these questions are foundational: the mandate presupposes that the policy rate exerts a measurable influence on the price level. A substantial empirical literature supports the proposition that an increase in the policy rate is, on average, associated with a subsequent decline in prices (Romer and Romer, 2004; Christiano, Eichenbaum, and Evans, 2005; Boivin, Giannoni, and Mihov, 2009; Aruoba and Drechsel, 2024). The estimated effects, however, are characterized by “long and variable lags”: considerable transmission delays and non-trivial uncertainty around the point estimates.

Standard textbook treatments offer several mechanisms consistent with this empirical regularity. A higher policy rate is expected to dampen aggregate demand, thereby exerting downward pressure on prices. In open-economy settings, it may further induce a nominal appreciation of the exchange rate, lowering the price of imported goods. These demand- and exchange-rate-based channels, however, are not exhaustive. The cost channel, where higher interest rates raise firms’ financing costs and place upward pressure on prices, operates in the opposite direction and complicates the conventional account of monetary transmission.

This study asks how monetary policy transmits to consumer prices. We examine the case of Norway, a small open economy with a floating exchange rate and a flexible inflation-targeting regime. Using monetary policy shocks from Holm, Paul, and Tischbirek (2021) as instruments for interest rate changes, we estimate their effects on sub-indices of the monthly Consumer Price Index over the period 2001–2018.

Our first exercise estimates the response of aggregate and average consumer prices to changes in the interest rate. The point estimate implies that prices grew by about 2% less after five years following a 100-basis-point increase, but the effect is delayed and imprecisely estimated: the 95 percent confidence band ranges from about 4% lower to 1% higher cumulative price growth, and the response becomes economically meaningful only after several years. The magnitude and timing of this aggregate response are consistent with prior evidence for Norway (Holm, Paul, and Tischbirek, 2021) and the United States (Aruoba and Drechsel, 2024).

Disaggregating to the level of individual CPI sub-indices reveals substantial heterogeneity. Some prices grow significantly more slowly when rates increase, most notably *accommodation services* and *food and non-alcoholic beverages*. Others grow significantly faster, such as *rentals for housing*. In contrast to the U.S. evidence (Aruoba and Drechsel, 2024),

we find little support for a strong common price response, suggesting that no single transmission channel dominates across categories. Instead, the dispersion points to monetary policy reaching different prices through different mechanisms.

To interpret this heterogeneity, we examine three transmission channels: the cost, aggregate demand, and exchange rate channels.

We begin with the cost channel (Barth and Ramey, 2001), for which the evidence is clearest. Higher interest rates raise firms' financing costs, particularly for those carrying substantial debt, and should therefore push prices upward most strongly in sectors where interest expenses constitute a large share of costs. We proxy firms' interest-rate exposure with long-term debt relative to total costs and net working capital relative to sales.¹ The evidence points to a meaningful role for this channel: sectors with higher average debt-to-cost ratios reduce price growth less in response to an interest rate increase. The association is statistically significant and economically sizeable. Moving from a debt-to-cost ratio of 0 to 1.4 (the CPI-weighted sample average) is associated with an additional 0.7-percentage-point increase in prices after four years, and the implied difference is several times larger for the high-debt rental sectors, where debt-to-cost ratios reach roughly 6. The result is driven primarily by sectors with very high debt shares of total costs, most notably rentals for housing, where prices rise rather than fall following a monetary tightening.

We next consider the aggregate demand channel, which predicts that a higher interest rate lowers demand and eventually induces most firms to lower price growth, producing a common factor across categories analogous to the U.S. results (Aruoba and Drechsel, 2024). Even at long horizons, however, the dispersion in price responses remains substantial in our data, suggesting that the aggregate demand channel operates more weakly in Norway than in the U.S.

To shed light on the aggregate demand channel, we group consumer prices into categories: durables, non-durables, and services. The evidence is arguably indirect, but the hypothesis is that for a small open economy like Norway, services and non-durables prices are determined mostly by Norwegian demand, while durables prices depend more on the exchange rate and world-market prices. We find that non-durable prices grow more slowly after an interest rate increase, whereas the response of durable prices is estimated imprecisely. Service prices first rise by around 1.4 percent after one year, and then revert. The service price response mirrors the response of hourly wages, which also rise in the first year before gradually declining, and total wages and hours worked fall. These results

¹The debt-to-cost ratio is a natural proxy in Norway because corporate debt consists predominantly of bank loans, around 90 percent of which carry adjustable rates set as a premium over the money market rate (Cao, Hegna, Holm, Juelsrud, König, and Riiser, 2025). The results are similar if we use interest costs as a share of total costs instead.

are consistent with an aggregate demand channel operating through the labor market, but the evidence is indirect, and the estimates are imprecise.

Norway's wage-setting institutions offer one candidate explanation for why the aggregate demand channel may be weaker in Norway than in the United States. Under the country's system of pattern bargaining (Aukrust, 1977), nationwide wage growth is anchored to what the export sector can sustain and, to a lesser extent, determined by domestic conditions. This arrangement is designed to insulate domestic wages from fluctuations in domestic demand, so that a contraction in demand should pass through to wages more slowly and more weakly than in economies with decentralized wage setting. The gradual, lagged decline in hourly wages we document is what such a mechanism would predict.

Finally, we turn to the exchange rate channel. Standard theory predicts that a surprise increase in the interest rate generates an immediate appreciation, which can in turn lower the prices of imported goods. Consistent with prior evidence, we find that a higher interest rate appreciates the Norwegian krone, measured by the import-weighted I-44 index: a 100-basis-point increase produces a roughly 6% appreciation after about 12 months.

Whether this appreciation passes through to consumer prices is a separate question. To gauge the strength of the exchange rate channel, we estimate how price responses to monetary policy vary with import exposure, measured both as imported inputs embodied in domestic production and as directly imported final consumption. Indices with higher import content tend to grow more slowly following an interest rate increase, in line with the predicted mechanism, but the estimates are imprecise, and the evidence remains inconclusive.

In sum, the paper delivers three messages. First, a higher interest rate tends to lower price growth, but with a delay and considerable imprecision. Second, the average response hides substantial and persistent dispersion across sectors. Third, among the three channels we examine, the cost channel emerges most clearly: following an interest rate increase, prices grow faster in sectors with substantial debt burdens, particularly housing rentals. The evidence is also consistent with an aggregate demand channel, albeit one that appears weaker than in the United States, and with an exchange rate channel, albeit one that is imprecisely identified.

Related literature. This paper contributes to three strands of literature.

A growing literature investigates how monetary policy affects disaggregated price indices. Some early contributors to this literature have mostly focused on how monetary policy affects prices, depending on the relative price stickiness of the sectors (Bils, Klenow,

Kryvtsov, et al., 2003; Boivin, Giannoni, and Mihov, 2009) or on exploring variations in channels across sectors (Barth and Ramey, 2001; Ravenna and Walsh, 2006; Balke and Wynne, 2007; Baumeister, Liu, and Mumtaz, 2013). The recent inflation episode has led to a renewed interest in this topic, spurring a literature exploring heterogeneity in price responses to monetary policy more broadly (Borio, Disyatat, Xia, and Zakrajšek, 2021; Luo and Villar, 2023; Guerrieri, Marcussen, Reichlin, and Tenreyro, 2023; Buda, Carvalho, Corsetti, Duarte, Hansen, Moura, Ortiz, Rodrigo, Rodríguez Mora, and Alves da Silva, 2025; Arnaut and Bengali, 2024; Aruoba and Drechsel, 2024). Our contribution to this literature is to examine how prices respond to monetary policy in a small open economy where we expect monetary transmission channels to operate differently. For example, in contrast to Aruoba and Drechsel (2024), our evidence suggests that there is less of a common price response across sectors, and thus that monetary transmission is less dominated by one common monetary transmission channel.

Our paper relates specifically to the evidence on the cost channel of monetary policy. Barth and Ramey (2001) present evidence suggesting the existence of a cost channel because in many industries output falls while price-wage ratios rise after a contractionary monetary policy shock. Similarly, Ravenna and Walsh (2006) construct and estimate a New Keynesian model where interest rate changes directly affect firms' debt service costs. On the empirical side, Gaiotti and Secchi (2006) document pricing behavior consistent with a cost channel in a large panel of Italian firms. The recent inflation episode has renewed interest in this channel: using euro-area firm-level data, Albertazzi, Ferrando, Gori, and Rariga (2025) and Core, De Marco, Eisert, and Schepens (2025) find that firms with greater reliance on debt and working capital raise prices following monetary tightening. We contribute to this literature by providing direct evidence of a cost channel of monetary policy where prices tend to rise in response to an interest rate increase in sectors with a high debt share relative to total costs.

Our paper also relates to the literature on monetary transmission to prices in Norway. Several papers examine how monetary policy affects the aggregate consumer price index (see, e.g., Holm, Paul, and Tischbirek, 2021), or specific prices such as the exchange rate (Bjørnland, 2008; Holm, Paul, and Tischbirek, 2021; Bjørnland, Brubakk, and Maffei-Faccioli, 2024) or house and stock prices (Bjørnland and Jacobsen, 2010, 2013; Robstad, 2018; Larsen, 2018; Rosenberg, 2019; Holm, Paul, and Tischbirek, 2021). We contribute to this literature by estimating the price response of disaggregated price series to monetary policy, and by exploring the role of several monetary transmission channels to prices in Norway.

2 Data

Prices. Statistics Norway classifies price indices following the United Nations Statistics Division’s *Classification of Individual Consumption according to Purpose* (COICOP). The analysis in this paper is at the 3-digit level. Table A.1 in Appendix A lists all price indices. Measuring prices in an economy presents several challenges that may impact the interpretation of our results. We discuss these challenges and how we address them in this section.

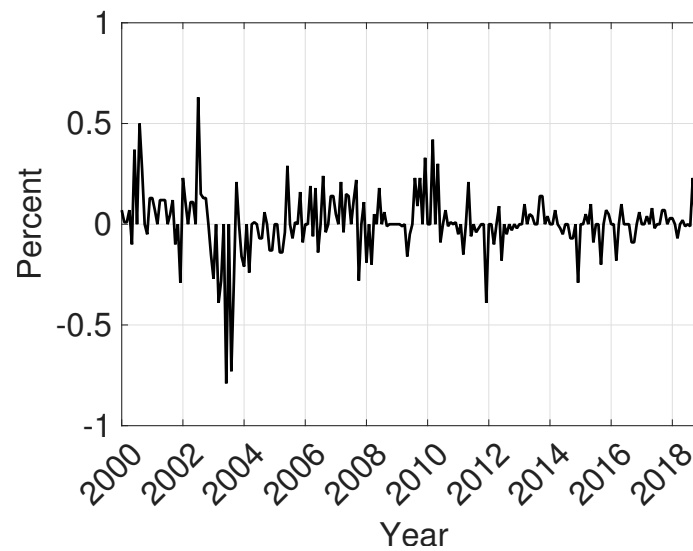
First, there is dispersion in the extent to which prices are measured precisely. Some prices, such as “food and beverages,” are calculated using vendor scanner data. These are measured precisely as they are derived from transactions across the entire product dimension. Other prices are calculated from samples of representative goods and vendors. For example, prices in “restaurants and accommodation” are obtained from repeatedly asking a selected group of restaurants and hotels. Second, quality changes represent an ever-recurring challenge when measuring price changes. The price index is typically designed to capture the price movements in the same basket of goods over time. Prices for a good may change either because the price moves or because there are quality improvements. For these first two challenges, we rely on the data from Statistics Norway in our analysis. To the extent possible, Statistics Norway deals with these challenges according to the common European statistical framework (EUROSTAT).

Third, some sectors are partly or wholly government-regulated. For example, transport services are typically regulated by the municipality, and the public provides a large part of education and health services at no cost. We remove prices in sectors that are primarily government-regulated from our sample. This includes 044 *Water supply and miscellaneous services relating to the dwelling*, 045 *Electricity, gas and other fuels*, 061 *Medical products, appliances and equipment*, 062 *Out-patient services*, 073 *Transport services*, 081 *Postal services*, and 124 *Social Protection*. Table A.1 in Appendix A details how we define COICOP codes as government-regulated, durables, non-durables, and services. Care should be taken when comparing our results to results in countries where costs related to health and education constitute a significant fraction of expenditures.

Our final price sample comprises 28 sub-indices, covering approximately 83% of the Consumer Price Index between January 2001 and December 2018. When estimating impulse responses of the policy rate, aggregate price indices, and national accounts data (including wages and hours), our sample spans from January 1995 to December 2018.

Monetary policy. We are concerned with how interest rate changes by the central bank affect prices. To represent the interest rate, we use the key policy rate at the end of

the month. Because the central bank responds to economic conditions when setting the interest rate, interest rate changes are endogenous. To address this endogeneity issue, we use the monetary policy shocks from [Holm, Paul, and Tischbirek \(2021\)](#) as instruments for unexpected monetary policy innovations. [Holm, Paul, and Tischbirek \(2021\)](#) construct monetary policy shocks following the approach in [Romer and Romer \(2004\)](#) as the change in the interest rate that is unrelated to Norges Bank's forecasts about output and inflation. The original series spans from 1994 to 2018 and is constructed at the frequency of monetary policy meetings. We use the series at the monthly frequency by summing up all shocks within the month. We refer to [Holm, Paul, and Tischbirek \(2021\)](#) for the estimation of the shocks and the tests for the exogeneity of the shock series.



Notes: The figure displays the monetary policy shocks from [Holm, Paul, and Tischbirek \(2021\)](#).

Figure 1: Monetary policy shocks.

Figure 1 presents the monthly shock series. The main event is a significant contraction in 2002 (positive shocks), followed by a series of expansionary shocks in 2003 (negative shocks). Reassuringly, a committee of external experts evaluating Norges Bank criticized the interest rate decisions in 2002 and 2003 as policy mistakes, citing the 2002 rate as too contractionary and the 2003 rate as too expansionary ([Bjørnland, Ekei, Geraats, and Leitemo, 2004](#)). For further discussion and documentation of the properties of this shock series, including the unpredictability of the shocks from past aggregate data and an explanation of the events that led to the large identified shocks, we refer to [Holm, Paul, and Tischbirek \(2021\)](#).

Variables. Besides the price indices from Statistics Norway, we use a number of other variables in the analysis. First, we use the averages within a month of the I-44 (import-weighted) exchange rate from Norges Bank.

Second, we use several variables from the firm balance sheets and income statements in the accounts statistics of Statistics Norway. Specifically, we use the long-term debt (excl. debt within the same corporate group),² cost components (total, wages, materials, and interest expenses),³ sales, current assets (including inventories), current liabilities, and operating profits from the firm's accounts. We define all variables at the sectoral level (NACE) and then transform them to COICOP as described below. We define the debt-to-cost ratio as long-term debt divided by operating costs at the COICOP level. The net working capital-to-sales ratio is defined as current assets net of current liabilities divided by total sales at the COICOP level. We also compute the interest-expenses-to-cost and the interest coverage ratio (operating profits divided by interest expenses) for results reported in Appendix C. We sum all firm-level variables over firms and over 2008 to 2019 within each NACE sector before forming the ratios.

Third, we measure import exposure as the import content of household final consumption. We first compute the direct and indirect imported-input content of domestic production using the 2012 input-output table for Norway from the World Input-Output Database (WIOD, [Timmer, Dietzenbacher, Los, Stehrer, and de Vries, 2015](#)), following the approach in [Dhyne, Kikkawa, Mogstad, and Tintelnot \(2023, Section II.D\)](#). Specifically, the import content of domestic production is defined as the sum of the direct import share of the sector and the import shares of its suppliers, weighted by the supplier sectors' input shares. Since household consumption also includes direct imports of final goods and services from foreign sectors, we then calculate the sector-specific import content of household final consumption as the sum of: (1) the imported-input content of the domestic sector, weighted by the share of household final consumption sourced domestically, and (2) the share of household final consumption sourced directly from the foreign sector. The import content measure is defined at the NACE level and transformed to COICOP using the bridge matrix described below.

Fourth, we use quarterly aggregate wages and hours from the national accounts when we estimate the impact of monetary policy on labor market outcomes.

²Long-term debt excludes debt to firms within the same corporate group, since the terms of intra-group loans are typically not tied to market rates.

³As with long-term debt, we exclude interest expenses within the same corporate group when we define interest expenses.

Mapping from NACE to COICOP classifications. The firm-level and trade data are organized by production-side NACE sectors, whereas the price indices follow the consumption-side COICOP classification. We map the former to the latter using a bridge matrix developed for Norway by [Cai and Vandyck \(2020\)](#), which yields measures of debt, costs, current assets and liabilities, sales, and import content for each 3-digit COICOP price index. We describe the mapping in detail in Appendix [A](#).

3 Empirical strategy

We estimate the effects of monetary policy on prices using instrumental variable local projections. We rely on a baseline specification to trace the response of aggregate and individual prices (Section [4](#)) and on an interaction specification to relate these responses to sector characteristics when we examine transmission channels (Section [5](#)).

3.1 Baseline specification

We estimate the effect of monetary policy on prices using an instrumental variable local projection framework:

$$p_{i,t+h} - p_{i,t-1} = \alpha_{i,h} + \beta_{i,h,0}^r \Delta r_t + \sum_{s=1}^{36} \beta_{i,h,s}^r \Delta r_{t-s} + \sum_{s=1}^{11} \beta_{i,h,s}^p \Delta p_{i,t-s} + \beta_m + \varepsilon_{i,t,h}, \quad (1)$$

$$\Delta r_t = \delta_0 + \delta_1 Z_t + \sum_{s=1}^{36} \delta_{1,s} \Delta r_{t-s} + \delta_m + u_{i,t} \quad (2)$$

where p_i is the log price of sector i , Δr_t is the change in the nominal key policy rate from beginning until end of month t , Z_t is the monetary policy shock, $\alpha_{i,h}$ is a horizon-fixed effect for sector i , β_m and δ_m are month-fixed effects, and $\varepsilon_{i,t,h}$ and $u_{i,t}$ are error terms. We use the predicted interest rate change $\widehat{\Delta r_t}$ from equation (2) when we estimate equation (1). The specification includes three years of interest rate lags and one year of price change lags. When we estimate the system on aggregate data, the price controls are also included in the first-stage regression. When we estimate using quarterly data, we include the number of lags corresponding to the same time length as for the monthly controls. We use [Newey and West \(1987\)](#) standard errors when estimating the regression system using only time-series data and [Driscoll and Kraay \(1998\)](#) standard errors when estimating the system with disaggregated price index data. We weight the regressions by CPI expenditure shares in

specifications using the disaggregated price-index panel.⁴

Interpretation. The dependent variable in equation (1) is the cumulative change in the log price index from month $t - 1$ to month $t + h$. The coefficient $\beta_{i,h,0}^r$ therefore measures by how much cumulative price growth over $h + 1$ months differs from what it would have been absent the interest rate change. A negative estimate means that prices grow more slowly than they otherwise would have. We therefore interpret negative β 's as "prices grow by less" and positive β 's as "prices grow by more" following an interest rate increase. Because the outcomes are aggregate price indices, the estimates capture the combined effect of all mechanisms triggered by the interest rate change, including any changes in supply, and we cannot attribute them to a single mechanism.

3.2 Interaction specification

In several of the settings below, we estimate the association between the effect of monetary policy on prices and the variation in a variable X_i (e.g., import content, debt-to-total costs, or the net working capital-to-sales ratio). In these cases, we estimate the interaction effect between monetary policy and the variable X_i using the following instrumental variable local projection framework:

$$p_{i,t+h} - p_{i,t-1} = \alpha_{i,h} + \beta_{i,h,0}^r \Delta r_t + \sum_{s=1}^{36} \beta_{i,h,s}^r \Delta r_{t-s} + \beta_{i,h}^{\text{int}} \Delta r_t \cdot X_i + \sum_{j=1}^{11} \beta_{i,h,j}^p \Delta p_{i,t-j} + \beta_m + \varepsilon_{i,t,h}, \quad (3)$$

$$\Delta r_t = \delta_0 + \delta_1 Z_t + \delta_2 Z_t \cdot X_i + \delta_3 X_i + \sum_{s=1}^{36} \delta_{1,s} \Delta r_{t-s} + \delta_m + u_{i,t}, \quad (4)$$

$$\Delta r_t \cdot X_i = \delta_0^{\text{int}} + \delta_1^{\text{int}} Z_t + \delta_2^{\text{int}} Z_t \cdot X_i + \delta_3^{\text{int}} X_i + \sum_{s=1}^{36} \delta_{1,s}^{\text{int}} \Delta r_{t-s} + \delta_m^{\text{int}} + u_{i,t}^{\text{int}} \quad (5)$$

where p_i is the log price of sector i , Δr_t is the change in the nominal key policy rate from beginning until end of month t , X_i is the interaction variable of interest (constant for sector i), Z_t is the monetary policy shock, $\alpha_{i,h}$ is a horizon-fixed effect for sector i , β_m , δ_m , and δ_m^{int} are month-fixed effects, and $\varepsilon_{i,t,h}$, $u_{i,t}$, and $u_{i,t}^{\text{int}}$ are error terms. We use the predicted interest rate change $\widehat{\Delta r_t}$ and interaction term $\widehat{\Delta r_t} \cdot X_i$ from equations (4) and (5) when we

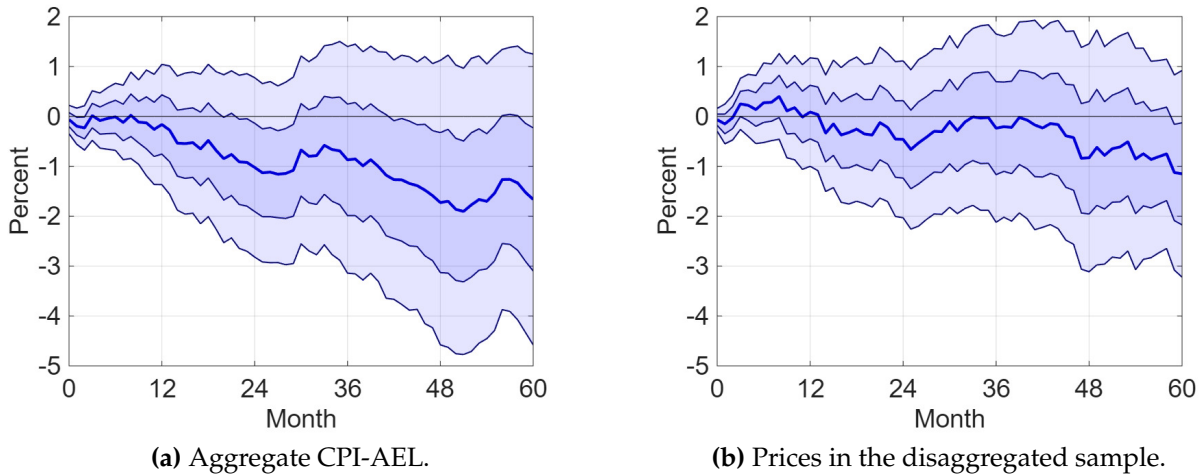
⁴The monetary policy shock is a strong instrument. In the first stage of the aggregate CPI-AEL regression, the (Kleibergen-Paap) F -statistic, computed with the same autocorrelation correction as the standard errors, is 83. In the CPI-weighted panel, it is 88 at horizon 0 and 74 at horizon 60. In the interaction specifications of section 5, which have two endogenous regressors, the F -statistics are between 37 and 44. All are well above the Stock-Yogo critical values for weak instruments.

estimate equation (3). As before, the specification includes three years of interest rate lags and one year of price change lags. We use [Driscoll and Kraay \(1998\)](#) standard errors when estimating the regression system using the disaggregated price-index panel, and weight the regressions by CPI expenditure shares.

Interpretation. The dependent variable in equation (3) is the cumulative change in the log price index from month $t-1$ to month $t+h$. The coefficient $\beta_{i,h}^{\text{int}}$ measures how much the relative cumulative price growth response to the interest rate change over $h+1$ months is associated with the interaction variable X_i . We therefore interpret negative β^{int} 's as “variable X_i is associated with slower price growth following an interest rate increase” and positive β^{int} 's as “variable X_i is associated with faster price growth following an interest rate increase.”

4 Price responses to monetary policy

4.1 Aggregate and average price responses



Notes: The figures show the response of the CPI-AEL index and the CPI-weighted average response across the disaggregated price indices to a 100-basis-point interest rate increase, estimated using (1) and (2). 95 and 68 percent confidence bands are shown, computed using [Newey and West \(1987\)](#) standard errors in (a) and [Driscoll and Kraay \(1998\)](#) standard errors in (b).

Figure 2: Aggregate and average price responses to monetary policy.

The first result concerns the aggregate price index. Figure 2a presents the response of the consumer price index excluding electricity prices (CPI-AEL).⁵ There are three main takeaways. First, aggregate prices tend to grow less in response to a contractionary monetary policy shock. The effect is potentially sizeable: the point estimate suggests prices are almost 2 percent lower after five years than they otherwise would have been in response to a 100-basis-point increase in interest rates. Second, the price response to monetary policy is delayed, gradually building up over time. Third, the price response is uncertain. While there is a tendency for reduced price growth, the response is not statistically significant at any conventional level of significance.

Figure 2b shows the CPI-weighted average price response estimated across the disaggregated price indices in our sample. This sample is the 28-index baseline sample described in Section 2, which excludes the government-regulated indices, whereas the CPI-AEL in Figure 2a includes them. The response is similar to the aggregate CPI-AEL response, but somewhat smaller: the point estimate builds up gradually, so that prices have grown by about 1 percent less after five years, but it remains uncertain.⁶

The estimated responses in Figure 2a differ from the price response estimated in Holm, Paul, and Tischbirek (2021). The primary reason for this difference is that we employ an instrumental variable setup in the current paper, whereas Holm, Paul, and Tischbirek (2021) estimate the reduced-form effect of the monetary policy shock. The estimation using the instrumental variable raises the standard errors somewhat. In Figure B.2 of Appendix B, we estimate a similar regression to that of Holm, Paul, and Tischbirek (2021) in our sample. The estimated coefficients are similar, but the standard errors are smaller.

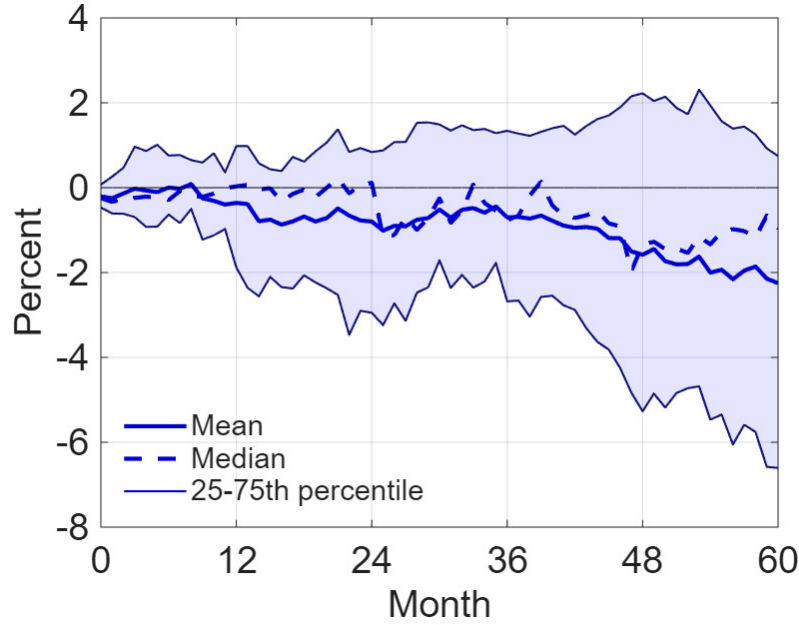
4.2 Disaggregated price responses

The average price response we estimate in Figure 2b is the weighted average of underlying price responses and may hide significant dispersion. To explore this dispersion, we estimate how prices at the 3-digit COICOP level respond to monetary policy.

Figure 3 displays the mean, median, and the 25th and 75th percentiles of the estimated

⁵We use CPI-AEL, the Consumer Price Index excluding electricity prices, because it has existed since 1995. In contrast, CPI-ATE, the conventional core inflation index in Norway, has existed only since 2002. In Figure B.1 in Appendix B, we show the estimated response of the policy rate to the monetary policy shock estimated using equations (1) and (2). The policy rate increases by 100 basis points and then continues to rise before eventually reverting after about two years.

⁶The difference between the two panels does not reflect sample composition: Figure B.3 in Appendix B shows that adding the government-regulated indices to the panel leaves the weighted average response virtually unchanged. Appendix B also reports a placebo exercise on the government-regulated indices themselves, whose prices should not respond to monetary policy: their weighted average response is small and never statistically significant (Figure B.4).



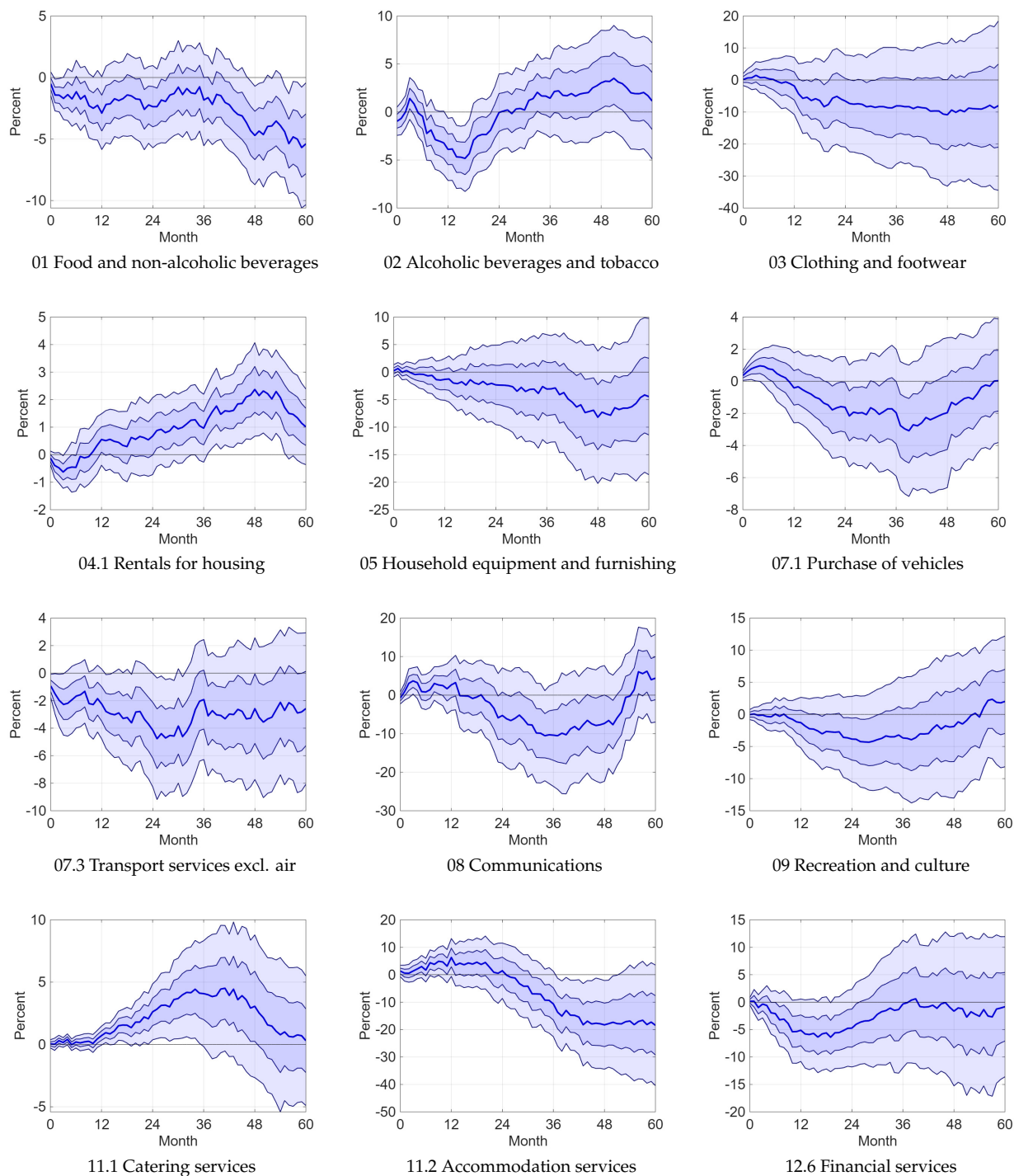
Notes: The figure shows the distribution of estimated individual price responses to a 100-basis-point interest rate increase, estimated using (1) and (2). All statistics are weighted using the CPI-weights.

Figure 3: The distribution of price responses to monetary policy.

price responses when estimating equations (1) and (2) on individual price indices.⁷ After five years the weighted median response is around -1 percent, close to the weighted-average price-index response in Figure 2b, while the weighted mean is around -2 percent, pulled down by a few sectors with sharply lower price growth. However, the mean and median responses hide significant heterogeneity. The shaded area in Figure 3 illustrates the dispersion in price responses, covering the 25th to 75th percentile of price responses. There is significant dispersion at all horizons. For example, after 60 months, the 25th percentile response is about -6 percent while the 75th percentile is around 1 percent.

Figure 4 further displays the price response of selected sectors. Some sectors exhibit significantly positive price responses to interest rate increases. For example, prices in *rentals for housing* grow by about 2.4 percent more at the peak following a 100-basis-point interest rate increase. In contrast, several prices grow significantly more slowly in response to an interest rate hike, such as *food and non-alcoholic beverages* and *accommodation services*. Our next section examines the factors that explain the variation in price responses, shedding light on the potential transmission channels at work.

⁷When we compute the mean, median, 25th and 75th percentiles, we weight each statistic using CPI-weights, applying the code in Shrestha (2025).



Notes: The figure shows the response of selected price indices to a 100-basis-point interest rate increase. 95 and 68 percent confidence bands are shown, computed using Newey and West (1987) standard errors.

Figure 4: Disaggregated price responses to monetary policy within selected COICOP categories.

5 Monetary transmission channels

We now ask which mechanisms drive the price responses documented above. We examine three channels: the cost, aggregate demand, and exchange rate channels. Table 1 summarizes, for each channel, the mechanism through which it operates, the sign of the price response it predicts, and the variation in the data we use to identify it. We present the channels in order of how clearly the data speak to them, beginning with the cost channel, for which the evidence is sharpest, and ending with the exchange rate channel.

Channel	Mechanism	Predicted price response	Identifying variation
Cost	Higher interest rates raise firms' financing costs.	Higher price growth, especially where interest expenses are a large share of costs.	Debt-to-cost and net-working-capital-to-sales ratios.
Aggregate demand	Higher interest rates reduce demand for consumption and investment.	Lower price growth, common across sectors.	Durables vs. non-durables vs. services, wages and hours.
Exchange rate	Higher interest rates lead to a krone appreciation.	Lower price growth, especially for import-intensive sectors.	Import content of consumption.

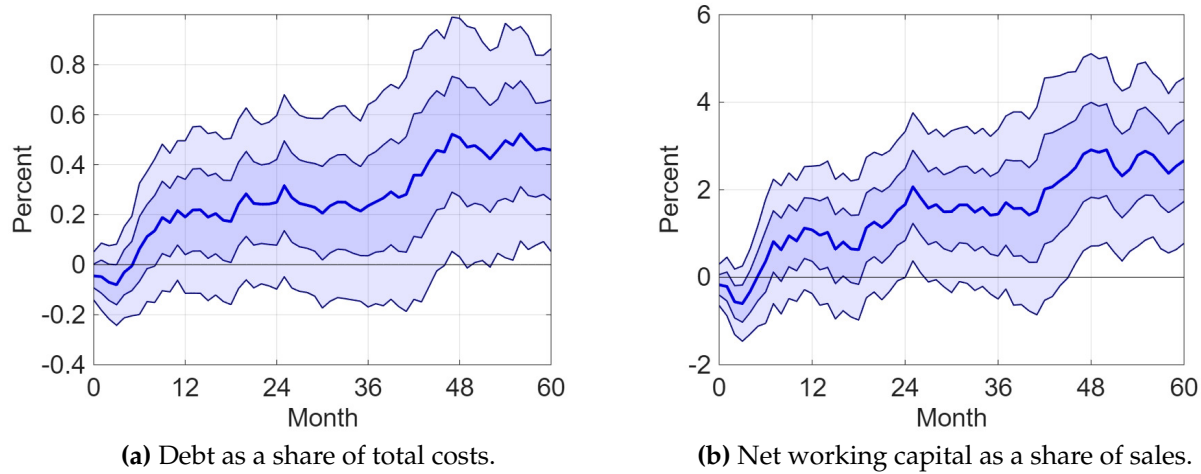
Table 1: Monetary transmission channels: mechanisms, predictions, and identification.

5.1 The cost channel

The first channel we examine is the cost channel of monetary policy (see, e.g., Barth and Ramey, 2001). An increase in the interest rate results in higher costs for firms, which may lead to higher prices. The cost channel thus operates in the opposite direction of the demand and exchange rate channels because a higher interest rate through the cost channel will result in higher prices.

The disaggregated price responses in Figure 4 already suggest that a cost channel may be responsible for some of the price increases in response to the interest rate increase. Specifically, prices for rentals for housing increase in response to an interest rate increase, a sector in which interest rate expenses are a relevant cost component.

To more rigorously explore the prevalence of a cost channel, we estimate how the price response to monetary policy varies with proxies for the extent to which interest rate expenses are a critical cost component in a sector. We examine two measures that can be constructed from sectoral accounts data: long-term debt (to lenders outside the firm's



Notes: The figures show the effect on consumer prices of a 100-basis-point interest rate increase interacted with the variables ‘debt as a share of total costs’ and ‘net working capital as a share of sales,’ estimated using (3), (4), and (5). Debt is long-term debt excluding debt to firms in the same corporate group (Section 2). 95 and 68 percent confidence bands are shown, computed using [Driscoll and Kraay \(1998\)](#) standard errors.

Figure 5: The association between the price response to monetary policy and debt as a share of total costs (a) and the net working capital as a share of sales (b).

corporate group) as a share of total operating costs, and net working capital as a share of sales.

The first measure is motivated by the cost channel model of [Ravenna and Walsh \(2006\)](#), in which firms borrow to finance production and the interest on the borrowed funds enters marginal cost. In such a model, the share of interest expenses in total costs is the product of the interest rate and the ratio of interest-bearing debt to costs. The debt-to-cost ratio therefore measures how sensitive a sector’s total costs are to a given change in the interest rate: a ratio of 2 implies that a 100-basis-point increase in the rate on the debt raises costs by 2 percent if fully passed through. We use the debt stock as our main measure because it isolates the exposure to rate changes from the level of rates at the time of measurement.⁸ The second measure follows [Barth and Ramey \(2001\)](#): a high net working capital relative to sales indicates that the firm ties up funds in inventories and receivables in excess of what is financed by suppliers, and thus faces an opportunity or financing cost that moves with the interest rate.

Whether these two measures capture exposure to monetary policy depends on the

⁸In Appendix C, we report results with two alternative measures based on realized interest expenses: interest expenses as a share of total costs, and the interest coverage ratio (operating profits divided by interest expenses). Both measures confirm the results.

structure of corporate financing. In Norway, corporate investment is financed mainly by internal funds, bank debt, and equity, and around 90 percent of bank debt contracts carry adjustable rates set as a premium over the money market rate, so that the average interest rate on the outstanding debt of non-financial firms tracks the policy rate closely (Cao, Hegna, Holm, Juelsrud, König, and Riiser, 2025). A change in the policy rate therefore quickly passes through to interest expenses on existing debt, and debt levels are a close proxy for exposure to interest rates. This contrasts with economies where a large share of corporate debt is at fixed rates, in which case only firms with floating-rate debt are exposed (Core, De Marco, Eisert, and Schepens, 2025). For net working capital, Giannetti, Burkart, and Ellingsen (2011) document that trade credit terms are often fixed at the industry level and insensitive to interest rates. Our measure nets trade credit received (in current liabilities) against receivables and inventories (in current assets), capturing the working capital the firm must finance itself at market rates rather than the terms of trade credit contracts. We treat the net working capital results as supporting evidence.

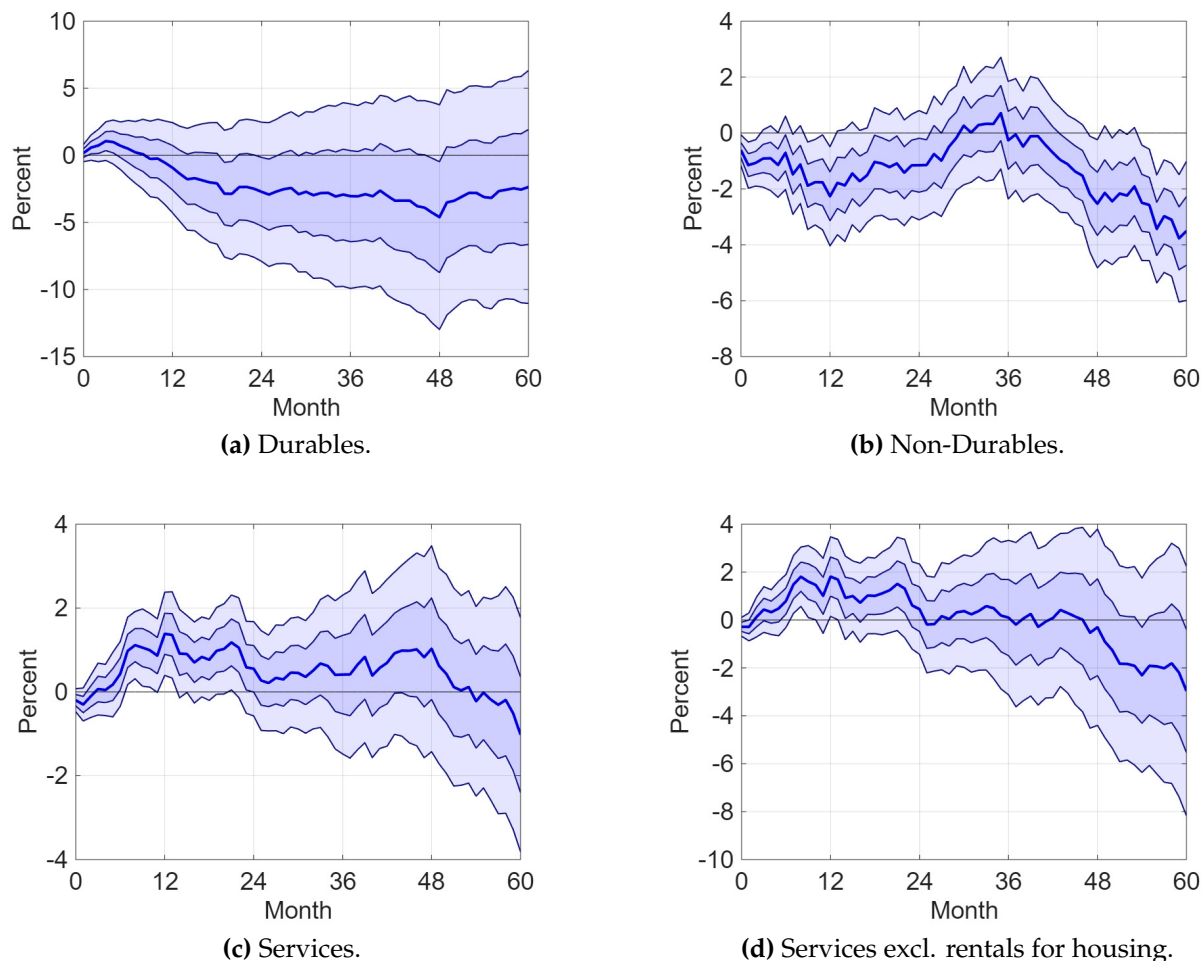
Figures 5a and 5b display the estimated interaction coefficients when estimating equations (3), (4), and (5) with the debt-to-total-cost and the net-working-capital-to-sales ratios. For both ratios, we estimate a positive interaction coefficient that gradually builds up and is statistically significant in years 4 and 5. Our results thus suggest that sectors with higher debt-to-total-cost or net working capital tend to raise prices more (or lower price growth less) in response to higher interest rates. The two interaction effects are also similar in magnitude when scaled by the variable's spread. Moving from a low- to a high-exposure sector (0 to 6 for debt-to-costs and from around 0 to 0.9 for working-capital-to-sales) is associated with about a 2.5-to-3 percentage-point change in the price response to monetary policy. Rentals for housing are a high-exposure sector by both measures, and important to explain the results.

5.2 The aggregate demand channel

The second channel we consider is the aggregate demand channel: in response to an interest rate increase, households and firms reduce their demand for consumption and investment, which may lead to lower prices.

The aggregate price response in Figure 2 and the distribution of price responses in Figure 3 both reveal a common pattern where price growth gradually decreases, pointing to an aggregate demand channel. However, the lack of statistical significance already suggests that this common channel is relatively weak.

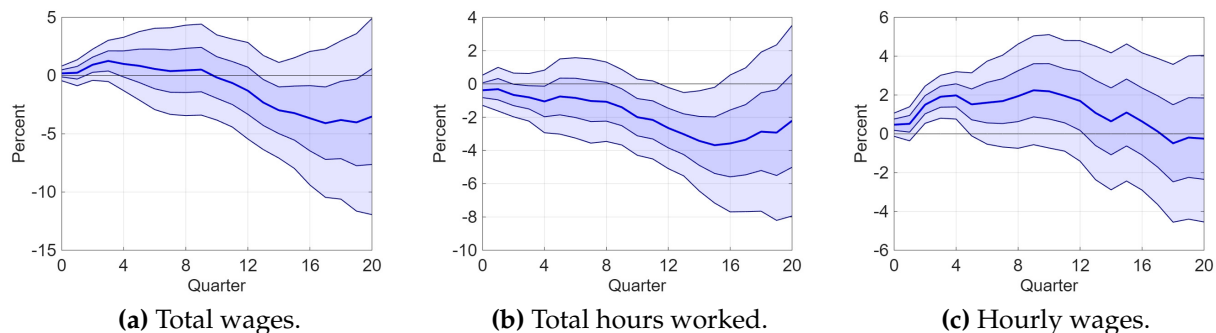
To understand the origin of this common response, we first categorize prices into three broad spending categories, as shown in Figure 6: durables, non-durables, and



Notes: The figures show the average effect on prices of durables, non-durables, services, and services excluding housing to a 100-basis-point interest rate increase, estimated using (1) and (2). 95 and 68 percent confidence bands are shown, computed using Driscoll and Kraay (1998) standard errors.

Figure 6: Price responses by consumption categories.

services. For durables, there is a tendency for price growth to slow, but it is not statistically significant. For non-durables, prices grow significantly more slowly in response to an increase in interest rates. For services, the response is not flat but hump-shaped: prices rise significantly in the first year, peaking at around 1.4 percent above the counterfactual path after 12 months, before reverting toward zero over the following years. Excluding actual and imputed rentals for housing from services leaves the short-run increase intact and reveals a tendency for service prices to grow more slowly than the counterfactual path at longer horizons, although the long-run estimates are imprecise. As we show next, the hump-shaped response of service prices closely tracks the response of hourly wages.



Notes: The figures show the effect on wages, hours worked, and hourly wages to a 100-basis-point interest rate increase, estimated using (1) and (2). 95 and 68 percent confidence bands are shown, computed using Newey and West (1987) standard errors.

Figure 7: Labor market responses to monetary policy.

A candidate explanation for the common pattern across non-durable goods and services is wage costs. The wage rate is an important input price, affecting prices across all sectors, especially those where wages are a large share of costs and can be passed through to prices. To investigate the role of wages in explaining the pattern, we estimate the total wage, total hours worked, and wage per hour responses to monetary policy using equations (1) and (2) with quarterly data from 1995 to 2018. Total wages decline by around 4% in year four in response to a 100-basis-point interest rate increase. However, while the total wage response may be informative, it could reflect either an increase in unemployment or a decline in actual wages. Indeed, in Figure 7b, we find that total hours worked decline by approximately 4% in year four in response to a 100-basis-point interest rate increase. Hence, the total wage response is driven almost entirely by the reduction in the number of hours worked.

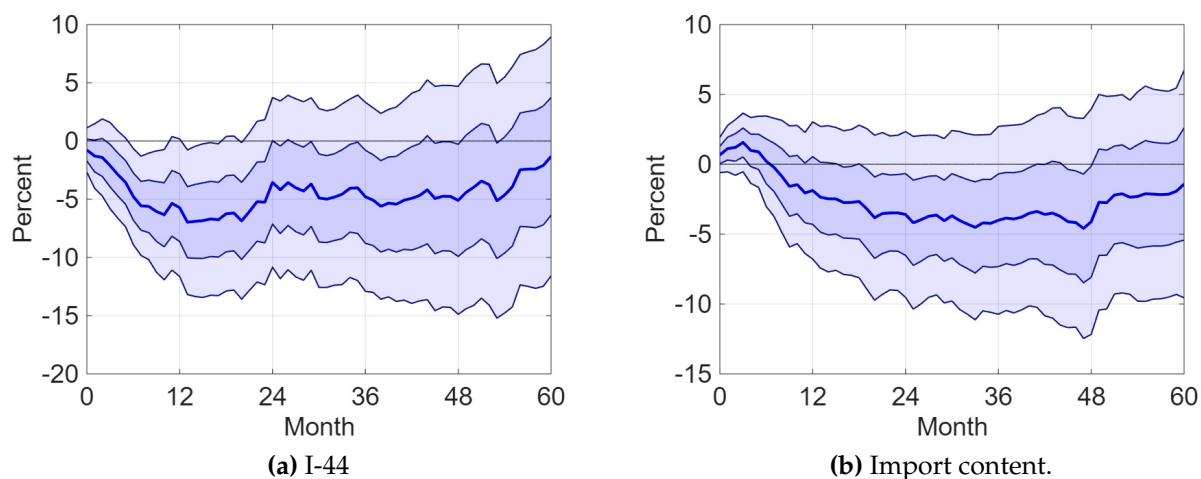
Figure 7c presents the estimated response of hourly wages in the national accounts to a 100-basis-point interest rate increase. The hourly wages, arguably a more precise measure of the wage costs (or the marginal costs) of existing firms, first increase, by around 2 percent in the first one to two years, then gradually decline. A composition effect likely explains part of the initial increase in hourly wages because hours tend to fall disproportionately in low-wage occupations during a recession. However, even with such a composition effect, hourly wages eventually stop increasing and may even decline. The timing of the hourly wage response matches the hump-shaped response of service prices in Figure 6c: service prices, for which wages are the dominant cost component, rise while hourly wages rise and revert as hourly wages revert. This co-movement is consistent with monetary policy transmitting to service prices through wage costs, although we cannot

distinguish a causal chain from wages to prices from a common response of both to the interest rate change.

Notably, if there is a decline in hourly wages in Figure 7c, it is gradual. It builds up over several years following monetary tightening, similar to the price responses. Hence, while we cannot confidently conclude that the aggregate demand channel plays a quantitatively significant role, we cannot rule out that it operates by slowing the growth of many prices with a substantial lag via the labor market.

5.3 The exchange rate channel

The third channel we consider is the exchange rate channel. When the central bank unexpectedly raises the nominal interest rate, several mechanisms suggest that the exchange rate may appreciate, which may eventually lead to lower prices for imported goods.



Notes: Figure (a) shows the response of the I-44 to a 100-basis-point interest rate increase, estimated using (1) and (2). Figure (b) shows the effect on consumer prices of a 100-basis-point interest rate increase interacted with the import content, estimated using (3), (4), and (5). 95 and 68 percent confidence bands are shown, computed using Newey and West (1987) standard errors in (a), and using Driscoll and Kraay (1998) standard errors in (b).

Figure 8: Exchange-rate responses and price responses by import content to monetary policy.

To explore the potential importance of the exchange rate channel, we first estimate the effect of monetary policy on the I-44 exchange rate (an import-weighted composite exchange rate constructed by Norges Bank) using equations (1) and (2). Figure 8a presents the results. A 100-basis-point increase in the interest rate tends to appreciate the exchange

rate (a negative response). The effect is quite sizable: the point estimate after one year is approximately a 6% exchange rate appreciation.

Even if the exchange rate appreciates in response to a higher interest rate, it does not necessarily imply that prices on imported goods will also decrease. We test the existence of the exchange rate channel by estimating how the price response to monetary policy varies with the import content of the price index using the interaction effect framework in equations (3), (4), and (5). The measure combines imported inputs embodied in domestically produced consumption with directly imported final consumption goods and services.

Figure 8b displays the estimated interaction effect for the import content. Consistent with an exchange rate channel, the estimated interaction coefficient is negative: price indices with higher import content tend to grow more slowly in response to a higher interest rate. However, the estimated coefficients are imprecise. Hence, while our results are qualitatively consistent with the presence of an exchange rate channel, we cannot confidently conclude that it is quantitatively important.

6 Conclusion

In this paper, we estimate the responses of disaggregated consumer prices to an increase in the interest rate. There is a tendency for price growth to slow, but the response is delayed and uncertain. The price responses exhibit wide dispersion, with some prices growing significantly more slowly while others grow faster in response to an interest rate increase. We examine three transmission channels: the cost, aggregate demand, and exchange rate channels. The clearest evidence is for the cost channel: prices tend to rise in response to an interest rate increase in sectors with high long-term debt relative to total costs, most prominently *rentals for housing*. The evidence is consistent with an aggregate demand channel operating slowly through wages, and with an exchange rate channel, but the estimates for these two channels are imprecise, and we cannot make confident claims about their quantitative importance.

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Online Appendix for Monetary Transmission to Prices: Disaggregated Evidence from Norway

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September 2026

A Appendix to Section 2

Mapping from NACE to COICOP classifications. The sectoral data are organized according to the NACE classification, which categorizes economic activities. NACE is closely aligned with the Classification of Products by Activity (CPA) classification in Eurostat, which categorizes the goods and services generated by these activities. Although firms may produce multiple goods or services that span different CPA categories, we consider this issue minor, given the relatively high level of aggregation employed in our analysis. We map data from the production side of the economy, as defined by CPA, to the consumption side, as determined by COICOP, using a bridge matrix developed for Norway by [Cai and Vandyck \(2020\)](#), which was constructed using a detailed Eurostat correspondence table and adjusted using a matrix procedure (RAS) to ensure consistency with the Norwegian national accounts. The procedure is further detailed in [Cai and Rueda-Cantuche \(2019\)](#). This process provides measures of debt, costs, current assets and liabilities, sales, and import content by 3-digit COICOP price indices.

Table A.1: Price indexes (3-digit COICOP) and accompanying categories

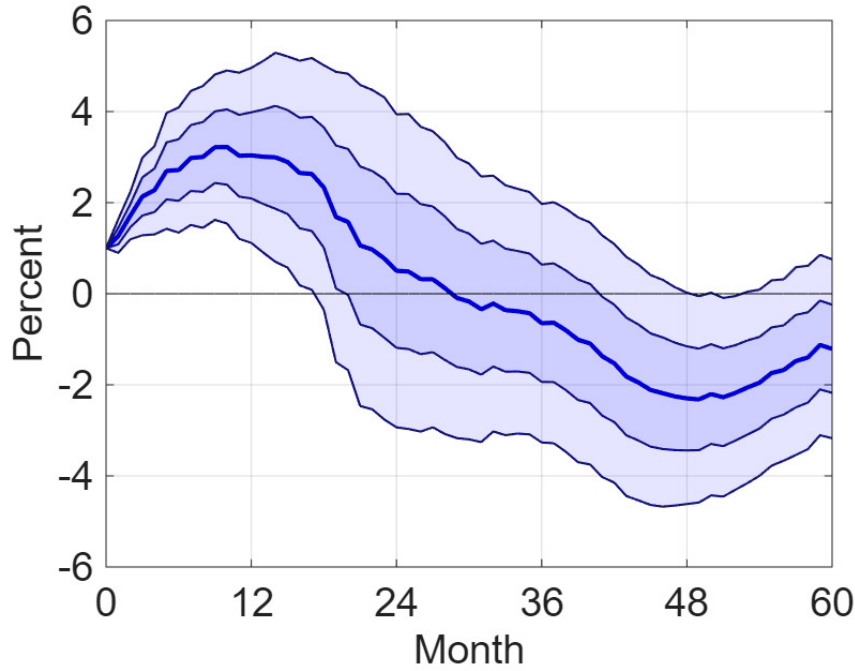
Price Index	Durables (D)	Non-Durables (ND)	Services (S)
011 Food		✓	
012 Non-alcoholic beverages		✓	
021 Alcoholic beverages		✓	
022 Tobacco		✓	
031 Clothing	✓		

Price Index	Durables (D)	Non-Durables (ND)	Services (S)
032 Footwear	✓		
041 Actual rentals for housing			✓
042 Imputed rentals for housing			✓
043 Maintenance and repair of the dwelling			✓
044 Water supply and miscellaneous services relating to the dwelling			✓
045 Electricity, gas and other fuels		✓	
051 Furniture and furnishings, carpets and other floor coverings	✓		
052 Household textiles	✓		
053 Household appliances	✓		
054 Glassware, tableware and household utensils	✓		
055 Tools and equipment for house and garden	✓		
056 Goods and services for routine household maintenance		✓	
061 Medical products, appliances and equipment		✓	
062 Out-patient services			✓
071 Purchase of vehicles	✓		
072 Operation of personal transport equipment			✓
073 Transport services			✓
081 Postal services			✓
082 Telephone equipment	✓		
083 Telephone services			✓
091 Audio-visual, photographic and information processing equipment	✓		
092 Other major durables for recreation and culture	✓		
093 Other recreational items and equipment, gardens and pets		✓	
094 Recreational and cultural services			✓
095 Newspapers, books and stationery		✓	
096 Package holidays			✓
111 Restaurant services			✓
112 Accommodation services			✓
121 Personal care	✓		
123 Personal effects nec	✓		
124 Social protection			✓
125 Insurance			✓
126 Financial services nec			✓
127 Other services nec			✓

Price Index	Durables (D)	Non-Durables (ND)	Services (S)
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Notes: The table lists all 3-digit COICOP indices. The baseline sample excludes primarily government-regulated indices: 044, 045, 061, 062, 073, 081, and 124.

B Appendix to Section 4



Notes: The figure displays the response of the policy rate to a 100-basis-point interest rate increase, estimated using (1) and (2). 95 and 68 percent confidence bands shown, computed using Newey and West (1987) standard errors.

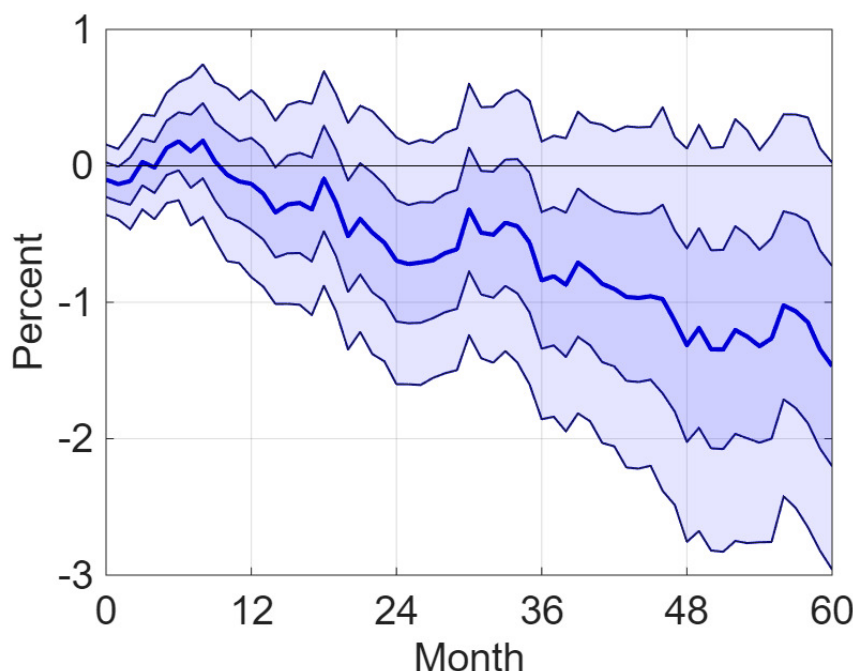
Figure B.1: The policy rate response to monetary policy.

In Figure B.2, we estimate the effect of the monetary shocks on the aggregate price index using the following regression specification:

$$p_{t+h} - p_{t-1} = \alpha_h + \beta_{h,0}^r Z_t + \sum_{s=1}^{36} \beta_{h,s}^r Z_{t-s} + \beta_m + \varepsilon_{t,h}. \quad (\text{B.1})$$

The main difference from the specification in the paper is that (B.1) estimates the reduced-

form impact of the monetary policy shock (normalized to a 100-basis-point increase) on prices, not the effect of a 100-basis-point interest rate change as in the other specifications. Moreover, this specification does not include lags of the log price changes because they were not included in [Holm, Paul, and Tischbirek \(2021\)](#).



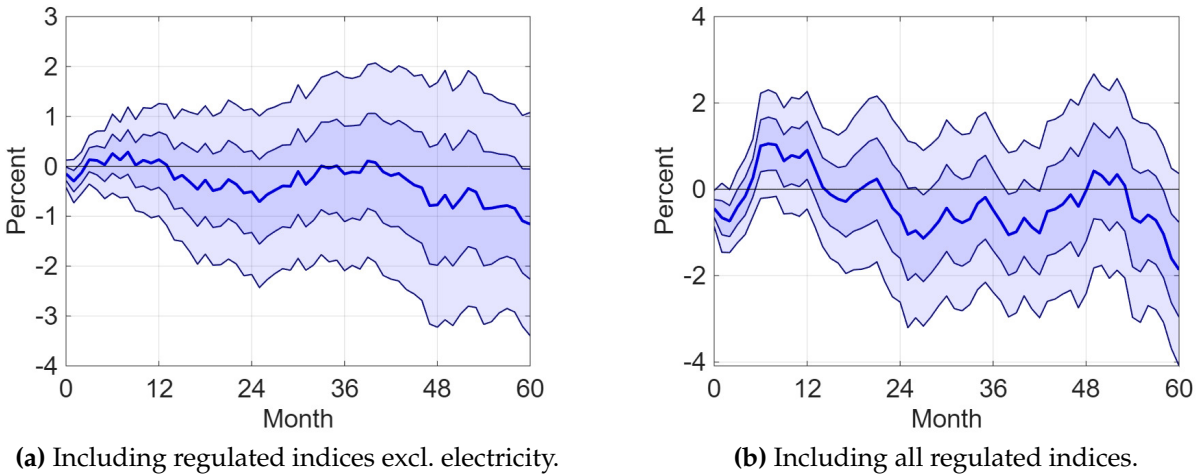
Notes: The figure displays the response of CPI-AEL to a 100-basis-point interest rate increase, estimated using (B.1). 95 and 68 percent confidence bands shown, computed using [Newey and West \(1987\)](#) standard errors.

Figure B.2: The aggregate price (CPI-AEL) response to monetary policy.

Government-regulated price indices. The baseline sample excludes seven government-regulated price indices (044, 045, 061, 062, 073, 081, and 124; see Table A.1). We ask whether their exclusion affects the average price response in Figure 2b, and whether these prices respond to monetary policy at all.

Average response including the regulated indices. Figure B.3 re-estimates the CPI-weighted average price response of Figure 2b on a panel that adds the government-regulated indices. Panel (a) adds the six regulated indices other than electricity, while panel (b) adds all seven. Adding the six regulated indices leaves the response virtually unchanged: the point estimate builds gradually to around -1.2 percent after five years, as

in the baseline, and remains statistically insignificant. Adding electricity, whose price is market-determined and highly volatile, makes the response noisier but does not change the picture at longer horizons. The exclusion of the regulated indices therefore does not explain the difference between the CPI-AEL response in Figure 2a and the weighted average of the disaggregated sample in Figure 2b.

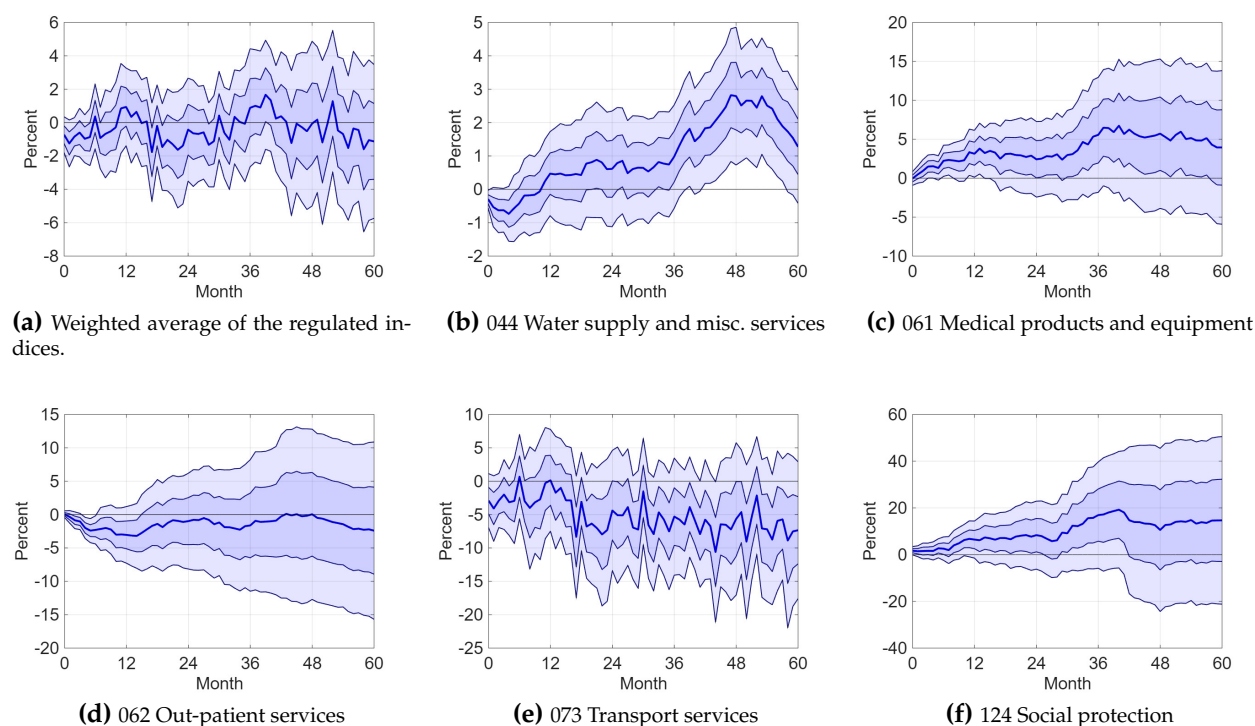


Notes: The figures show the CPI-weighted average response across the disaggregated price indices to a 100-basis-point interest rate increase, estimated using (1) and (2), when the government-regulated indices are added to the baseline sample of Figure 2b. Panel (a) adds 044, 061, 062, 073, 081, and 124, while panel (b) also adds 045 Electricity, gas and other fuels. 95 and 68 percent confidence bands are shown, computed using Driscoll and Kraay (1998) standard errors.

Figure B.3: Average price responses to monetary policy including the government-regulated indices.

Placebo: responses of the regulated indices. If prices in the regulated sectors are set administratively rather than in response to economic conditions, they should not respond to monetary policy. Figure B.4 reports the responses of the regulated indices other than electricity and postal services (081), which we exclude because the series is only available from July 2008, the same incomplete-coverage criterion applied to the baseline sample. Panel (a) shows the CPI-weighted average response of the remaining five indices, estimated with the panel specification (1) and (2). The point estimates are small relative to their standard errors and never statistically significant at conventional levels. Panels (b) to (f) show the individual responses, estimated series by series with Newey and West (1987) standard errors. Transport services (073) and social protection (124) show no significant response at any horizon. Out-patient services (062) show a significant decline only

around six months. Water supply and related services (044) rise by 2 to 3 percent four years after the interest rate increase. These municipal fees are set to cover costs, potentially including capital costs, so a delayed increase is not inconsistent with regulation and is, if anything, in line with the cost channel. Medical products (061) rise in the first year. This index combines regulated prescription drug prices with market-priced non-prescription products and appliances, so it is not a pure placebo. Overall, the placebo supports the identification strategy: the systematic responses documented in the main text are absent from prices that are not set in markets, and the exceptions have plausible institutional explanations.



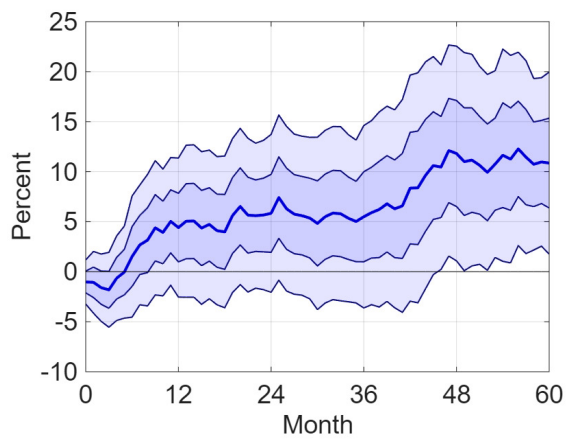
Notes: The figures show the response of the government-regulated price indices to a 100-basis-point interest rate increase, estimated using (1) and (2). 045 Electricity, gas and other fuels is excluded because it is market-priced, and 081 Postal services because the series is only available from July 2008. Panel (a) is the CPI-weighted average across the five remaining indices, with Driscoll and Kraay (1998) standard errors. Panels (b) to (f) are estimated series by series with Newey and West (1987) standard errors. 95 and 68 percent confidence bands are shown.

Figure B.4: Placebo: price responses of the government-regulated indices to monetary policy.

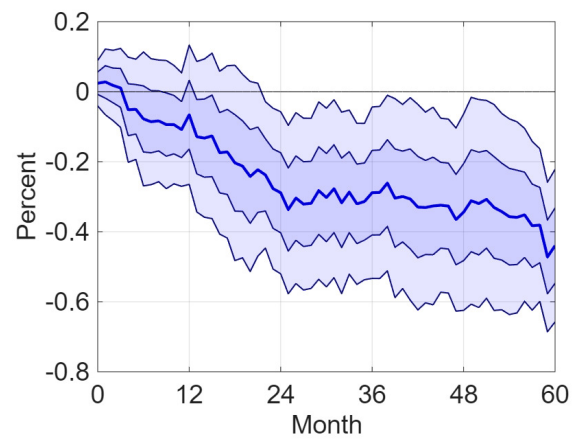
C Appendix to Section 5

The cost channel with interest-expense measures. The debt-to-cost ratio in Section 5.1 measures exposure to interest rate changes by the stock of long-term debt. A more direct measure of the burden of debt servicing is the share of interest expenses in total costs, and a common measure of a firm's headroom over that burden is the interest coverage ratio, the operating result divided by interest expenses. We construct both from the same sectoral accounts data as the debt-to-cost ratio, using interest expenses (excluding interest paid within corporate groups) and the operating result summed over firms and years within each NACE sector and bridged to COICOP as described in Appendix A. Across the indices in the baseline sample, the ratio of interest expenses to costs ranges from close to zero to about 0.25 for rentals for housing, and is almost perfectly correlated with the debt-to-cost ratio (correlation 0.999), because the implied average borrowing rate, between 4 and 6 percent, varies little across indices. The interest coverage ratio is a different variable: it is around 2 for rentals for housing and accommodation services, and between 3 and 14 for most goods and other services.

Figure C.1 reports the interaction estimates from equations (3), (4), and (5) with the two measures. Panel (a) uses interest expenses as a share of costs. As expected given the correlation, the response mirrors Figure 5a: the interaction coefficient builds up gradually and is statistically significant in years 4 and 5. Moving from an interest-expense share of 0 to 0.25, the value for rentals for housing, is associated with a price response that is about 2.7 percentage points higher after five years, the same magnitude as implied by the debt-to-cost ratio. Panel (b) uses the interest coverage ratio. The interaction coefficient is negative, meaning that sectors with more headroom over their interest burden lower price growth by more, and it is statistically significant from about two years onward. Moving from a coverage ratio of 2, as for rentals, to 6, a typical value for goods, is associated with a price response that is about 1.8 percentage points lower after five years. The interest coverage ratio is not available for insurance, which has no debt or interest expenses in the accounts data, so panel (b) uses one index fewer. Both measures thus confirm the cost channel result: prices rise, or grow faster, after an interest rate increase in sectors where interest expenses weigh heavily on costs and profits.



(a) Interest expenses as a share of total costs.



(b) Interest coverage ratio.

Notes: The figures show the effect on consumer prices of a 100-basis-point interest rate increase interacted with (a) interest expenses as a share of total costs and (b) the interest coverage ratio (operating result divided by interest expenses), estimated using (3), (4), and (5) on the baseline sample. Both measures are ratios of sums over firms and over 2008 to 2019 within NACE sectors, bridged to COICOP. 95 and 68 percent confidence bands are shown, computed using [Driscoll and Kraay \(1998\)](#) standard errors.

Figure C.1: The association between the price response to monetary policy and interest-expense measures of the cost channel.